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Hybrid spatial–temporal graph neural network for traffic forecasting

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ABSTRACT

Accurate traffic forecasting is the foundation of the intelligent transportation system (ITS). Among existing methods, researchers utilize deep neural networks to capture spatial–temporal correlations. While state-of-the-art methods have shown remarkable progress, they still have some limitations: (i) Most existing models struggle to make accurate prediction performance over longer time horizons due to the challenge of modeling heterogeneous temporal patterns. (ii) Recent studies still lack the ability to model the dynamic road network structure in a specific spatial–temporal context. To address these challenges, this work introduces a novel hybrid spatial–temporal graph neural network (HSTGNN) for traffic forecasting. In the temporal dimension, we design a decoupling layer to reveal the heterogeneous traffic patterns using frequency domain analysis. Moreover, we propose a dual-channel learner to capture complementary information from heterogeneous features, which enhances the model's capacity to handle complex temporal dynamics, facilitating long-term traffic forecasting tasks. In the spatial dimension, considering the road networks' inherent attributes and time-varying spatial structures, HSTGNN employs a novel hybrid graph learning approach to learn compound spatial correlations. This approach ensures that the model captures the comprehensive spatial correlations from both macro and micro perspectives. Finally, we conduct extensive experiments on six real-world traffic datasets, covering different traffic scenarios, to demonstrate the effectiveness of HSTGNN over state-of-the-art methods. The code is available at https://github.com/STGTraffic/HSTGNN.

1. Introduction

In recent years, with the rapid development of urban transportation, a large number of sensors have been deployed on the roads to detect traffic conditions. The collected data can be represented as time series as shown in Fig. 1. They can provide powerful data support for the smooth operation of urban transportation [1]. By analyzing historical data, researchers can gain insights into the distribution of traffic speeds and flows at different times along the road, which provides a foundation for traffic scheduling and signal control. However, accurate traffic forecasting is quite challenging due to the intricate temporal changes and dynamic spatial correlations.

Previous traffic forecasting research can be divided into two categories: knowledge-driven and data-driven methods. The knowledge-driven approach, such as queuing theory [2], simplifies user behavior but overlooks real-world traffic complexity. The data-driven approach often uses statistical methods such as Auto-Regressive Integrated Moving Average (ARIMA) [3] and Kalman Filtering (KF) [4] for simple time series prediction tasks. Nevertheless, traffic data is inherently non-stationary, with statistical properties such as mean and variance varying over time. As a result, these methods face significant challenges in accurately capturing the complex and dynamic temporal dependencies present in traffic data.

Recently, deep learning methods have been widely used in traffic forecasting due to their powerful data modeling capabilities. In the temporal dimension, Recurrent Neural Network (RNN) [5] and its variants, including Long Short-Term Memory (LSTM) [6] and Gated Recurrent Unit (GRU) [7], have been widely applied for time series modeling. The cyclic structure and parameter-sharing mechanism of RNNs ensure that the model efficiently captures and propagates the dynamics of time series data between time steps. In the spatial dimension, early researchers segmented the road network into grids and used convolutional neural network (CNN) [8] to extract spatial features. CNN applies shared convolutional kernels to aggregate features by performing weighted summation within the receptive fields. However, real-world road networks are inherently non-Euclidean, which prevents CNN-based methods from effectively extracting spatial dependencies. Since road networks can be naturally represented as graph structures, researchers have started using spatial–temporal graph neural networks (STGNNs) to extract features [9–11]. Compared to CNN, graph convolutional network (GCN) and graph attention network (GAT) can

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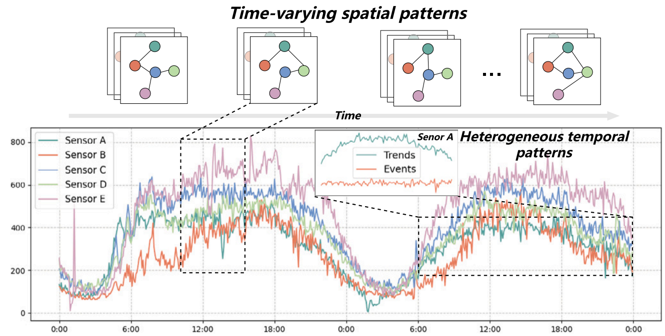

Fig. 1. Traffic series collected by road sensors. In the temporal dimension, traffic series show heterogeneous patterns, such as long-term trends and short-term fluctuations. In the spatial dimension, the strength of the relationship between nodes changes over time.

extract complex spatial correlations effectively, making them more efficient in traffic forecasting. In later studies, several methods have incorporated STGNNs with attention mechanisms to further improve model performance [12,13]. Despite the progress, there are still some limitations.

Firstly, in the temporal dimension, due to the heterogeneous temporal patterns within series, it is difficult for existing methods to distinguish their unique contributions to traffic dynamics, leading to sub-optimal performance in long-term forecasting. The heterogeneous temporal patterns refer to the diverse and varying characteristics within time series data. These patterns often have distinct frequencies, magnitudes, and durations. For example, Fig. 1 shows the traffic flow series collected by road senor A. It is evident that traffic series often comprises a variety of complex temporal components, such as trend components (low-frequency features, e.g., long-term growth trends) and random components (high-frequency features, e.g., short-term fluctuations due to random events). However, recent studies still tend to rely on a single unified structure to process all components of the time series [14–17]. The heterogeneous temporal nature of traffic data makes it challenging for these methods to model intricate temporal dynamics. This mismatch in modeling capability leads to a significant decline in performance with the forecasting horizons increasing, as models fail to adapt to the varying dynamics across different temporal components. Therefore, we argue that separating and modeling these components individually provides a clearer understanding of their distinct contributions to traffic dynamics. This approach also improves the model's ability to capture complex temporal dependencies effectively, facilitating long-term traffic forecasting tasks.

Secondly, in the spatial dimension, existing spatial–temporal neural networks lack the capacity to model complex and time-varying spatial correlations at specific traffic context. As Fig. 1 shows, the spatial topology relationship between the nodes can change over time. For example, during the morning and evening rush hours, the traffic flow between major roads increases, and the connection strength between nodes also increases. While during non-peak hours, these connections may weaken. Earlier studies used predefined matrices to reflect road topology and connection strength between nodes [18,19]. Factors like road segment distances and POI similarities were employed to define static spatial relationships in a fixed graph. Although these approaches are straightforward, they are unsuitable for scenarios where prior knowledge is unavailable. In subsequent studies, researchers started to build adaptive adjacency matrices by setting self-learning embedding vectors, achieving impressive effectiveness. Compared to predefined matrices, this approach no longer relies on prior knowledge, making it more suitable for diverse traffic conditions. However, either predefined or adaptive graphs are relatively static, failing to fully reflect the dynamic characteristics of the network over time [20,21]. Therefore, recent researchers have focused on constructing dynamic graphs to represent the evolving nature of traffic conditions [11,22,23]. By capturing the non-static evolution of node relationships, dynamic graphs encompass more spatial semantic information and improve the model's ability to represent complex spatial patterns.

To solve the above limitations, this paper introduces a novel **H**ybrid **S**patial-**T**emporal **G**raph **N**eural **N**etwork, namely HSTGNN, for traffic forecasting. To address the first limitation — the challenge of distinguishing and modeling heterogeneous temporal patterns, HSTGNN designs a decoupling layer and dual-channel learner. The decoupling layer is introduced to reveal the underlying stable and fluctuating components within the traffic series by frequency domain analysis. Furthermore, the dual-channel learner aims to capture complementary information from heterogeneous temporal patterns. These methods enable the model to learn the unique contributions of different temporal components, making it particularly advantageous for long-term forecasting tasks. To address the second limitation — the challenge of capturing time-varying spatial correlations, we propose a novel dynamic graph generation approach. This method integrates the dynamic traffic signals with spatial–temporal embeddings in a data-driven manner, enabling the model to dynamically extract spatial dependencies in a specific spatial–temporal context. Considering the inherent properties and time-varying spatial structure, we further incorporate it with the adaptive graph to model compound spatial dependencies. Finally, we demonstrate the superiority of HSTGNN on six real-world traffic datasets. Our main contributions can be summarized as follows:

- We introduce an effective spatial–temporal neural network HSTGNN for traffic prediction. In the temporal dimension, we design a decoupling layer to reveal heterogeneous traffic patterns and a dual-channel learner is further introduced for capturing complementary information from different temporal components.
- In the spatial dimension, we have devised a dynamic graph generation method that incorporates the traffic signals and spatial–temporal embeddings in a data-driven manner, which effectively models the time-varying properties of the road network. We further combine it with the adaptive graph learning algorithm to reflect the compound spatial correlations.
- We conduct extensive experiments on six real-world datasets to demonstrate our model outperforms state-of-the-art methods.

The remainder of the paper is as follows: Section 2 briefly reviews existing work on traffic forecasting. Section 3 presents the formalization of the problem and provides background knowledge. In Section 4, we describe the overall architecture of the model and introduce each module in detail. Furthermore, we compare our model with existing baseline models on six real-world datasets in Section 5. Finally, Section 6 summarizes our work in this paper.

2. Related work

In this section, we briefly review the early methods and existing spatial–temporal deep neural networks for traffic forecasting tasks.

2.1. Early studies for traffic forecasting

Traffic forecasting is a typical spatial–temporal series problem that uses historical data to predict future traffic conditions. Early researchers generally used traditional statistical methods, including HA (Historical Average) [24], ARIMA and KF, etc. Among these methods, ARIMA is widely used with many variants [25–27]. However, although the statistical methods have simple algorithms with easy implementation, they are unable to deal with complex non-linear traffic data. Machine learning methods were also widely used in the early stage. Dell et al. devised a time-aware multivariate nearest neighbors regression method for traffic flow prediction [28], and Luo et al. combined the discrete Fourier transform with SVR (Support Vector Regression) and proposed a hybrid prediction method [29]. Machine learning models often depend on feature engineering, which requires prior domain knowledge. Furthermore, their simplistic structures, limited parameter capacity, and low computational efficiency constrain their applications.

2.2. Spatial–temporal neural networks for traffic forecasting

In recent years, deep learning methods have gained widespread adoption due to their excellent data processing abilities and end-to-end predictions. Early studies typically used sequence-to-sequence models to extract temporal features while ignoring spatial features. For example, Duan et al. applied LSTM to predict future travel time [30]. Fu et al. applied LSTM and GRU to extract the temporal features of traffic series [31]. As the study evolved, researchers started to consider the spatial attributes of the road networks. CNNs were introduced to extract spatial features in the early stage. Zhang et al. proposed ST-ResNet for traffic forecasting, the model divided the road network into grids and utilized CNN-RNN combination to extract spatial–temporal dependencies [32]. Han et al. proposed a deep learning framework that incorporated a CNN-based deep clustering method to extract road shape features [33]. Wu et al. proposed a hybrid model combining CNN and two LSTMs [34], where the CNN was applied to capture spatial dependencies and the LSTMs were used to extract short-term temporal features and periodic patterns from past hours, days, and weeks. However, it is challenging for CNNs to represent the intricate structure of the traffic network due to the non-Euclidean nature of real-world road networks.

Recently, GNNs have received wide applications due to their modeling capabilities. GNNs have been widely applied in various domains, such as emotion recognition [35], social computation system [36], graph reinforcement and federated learning [37,38]. In traffic forecasting tasks, GNNs can better adapt intricate networks compared to CNNs, since roads can be naturally represented as nodes and their connections as edges in a graph. Zhao et al. proposed T-GCN for traffic speed forecasting [39]. They modeled the road network as a static graph and fed it into the GCN to capture spatial dependencies, GRU was employed to extract temporal features. Yu et al. designed STGCN that combined graph convolution with gated temporal convolution to capture spatial–temporal dependencies effectively [40]. Geng et al. proposed STMGCN which introduced a spatial–temporal multi-graph convolution framework to capture correlations between regions from three distinct spatial perspectives: spatial proximity, functional similarity, and regional connectivity [41]. This approach ensures a more comprehensive modeling of spatial dependencies. However, these methods heavily rely on manually predefined graph structures, which influence the performance of models. To address this issue, Wu et al. proposed GWNET to learn adaptive spatial correlation by setting learnable embedding vectors [10]. However, traffic series typically exhibit time-varying and dynamic spatial correlations. Therefore, modeling these correlations is essential for accurate traffic prediction. To address this challenge, Han et al. proposed DMSTGCN, which used three learnable matrices and a core tensor to learn the time-specific spatial dependencies of road segments, which aggregate hidden states of neighbor nodes to focal nodes by message-passing on the dynamic adjacency matrices [11]. However, assuming that traffic at a specific time of day shares the same graph can be limiting and may not accurately reflect the dynamic nature of traffic conditions. Guo et al. utilized the self-attention mechanism to compute the strength of dynamic spatial correlation between nodes and then adjusted the static weight matrix by an element-by-element dot product operation [42]. Although this approach considers dynamic spatial correlations, it still relies on the static matrix. Lan et al. introduced the DSTAGNN, generating a data-driven dynamic spatial–temporal graph by calculating spatial–temporal aware distances from traffic data [22]. Li et al. presented TPGCN, which integrates time-aware discrete graph structure estimation and dynamic personalized graph convolution components to model dynamic, heterogeneous dependencies in multivariate time series data [43]. Jin et al. proposed STUP, incorporating uncertainty quantification into traffic forecasting [44]. Shin et al. proposed the PGCN, which adapted graphs progressively during both training and testing phases to capture dynamic spatial–temporal dependencies in traffic forecasting [45]. Kong et al. proposed STPGNN, introducing the pivotal graph and TCN to capture complex spatial–temporal dependencies [17].

The attention mechanism was originally proposed in the field of natural language processing (NLP) to improve the quality of machine translation by dynamically focusing on the most relevant parts of an input sequence [46]. In subsequent studies, it was extended and applied to traffic prediction for capturing spatial and temporal dependencies. Guo et al. introduced the ASTGCN, which integrated spatial–temporal attention mechanisms and graph convolutions to capture complex spatial–temporal dependencies in traffic data [47]. Xue et al. introduced a method leveraging Fourier-series-based embeddings to model traffic periodicity and combined attention mechanism with a multi-hop diffusion process to capture large-scale structural dependencies [48]. Zhang et al. proposed STRGAT that leveraged a residual graph attention block for dynamic spatial feature aggregation and a sequence-to-sequence block for temporal dependencies modeling [49]. Tao et al. developed the MISTAGCN that integrated spatial and temporal attention mechanisms with graph and standard convolutions to comprehensively model traffic data, achieving superior performance on diverse datasets [50].

Compared with the above existing studies, our proposed HSTGNN has the following advantages and distinctions: (1) Heterogeneous temporal patterns modeling: HSTGNN introduces a Temporal Decoupling Layer (TDL) and Dual-Channel Learning (DCL) module to separately model heterogeneous temporal patterns, significantly improving adaptability to complex temporal dynamics in traffic data. (2) Compound spatial correlations modeling: The proposed dynamic graph enable the model capture dynamic spatial dependencies at specific spatial–temporal context. The Hybrid Graph Learning (HGL) module integrates adaptive and dynamic graph structures, enabling HSTGNN to model comprehensive spatial interactions in traffic networks.

3. Problem statement and preliminaries

This section will first introduce the definition of traffic forecasting, and then provide the background knowledge related to the GCN and GAT in this work.

3.1. Problem definition

Traffic Network: Given a road network can be viewed as a graph $G = (V, E, A)$, where V is the set of nodes, $|V| = N$ is the number of nodes, E denotes the set of edges, $A \in \mathbb{R}^{N\times N}$ indicates connectivity between nodes.

Traffic Series: The historical traffic series records on G at time slice t are denoted as a graph signal $X_t \in \mathbb{R}^{N\times F}$, where F is the dimension of each node's attributes.

Traffic forecasting: Traffic forecasting is a typical spatial–temporal prediction problem that gives previously observed traffic series in a road network to predict future traffic states. Given T historical graph signals $\mathcal{X} = \left[X_{(t-T+1)}, \dots, X_{(T)}\right] \in \mathbb{R}^{T\times N\times F}$, our goal is to learn a function F to forecast T' future graph signals $\hat{Y} = \left[\hat{X}_{(t+1)}, \dots, \hat{X}_{(t+T')}\right] \in \mathbb{R}^{T'\times N\times F}$ as accurately as possible. In brief, the forecasting process can be abstracted as Eq. (1):

$$\hat{Y} = F(X, A) \tag{1}$$

where X is the input graph signals, $\hat{Y}$ is the predicted signals and A denotes the adjacency matrix of the traffic network.

3.2. Graph neural networks

The GNN models have been successfully used in many applications, including document classification [51], unsupervised learning [52], and image classification [53], etc. At the same time, methods based on GCNs have become a focus for traffic forecasting. Indeed, constructing the adjacency matrix is crucial to the effectiveness of spatial modeling. However, most GCN-based approaches rely on adaptive or predefined graphs, which may not adequately capture the dynamic spatial correlations in the traffic network. In this paper, we propose a hybrid graph learning approach to learn optimal graph representations, and graph convolution and graph attention are applied to extract spatial features. We will explain the effectiveness of our architecture in Section 5.

3.2.1. Graph convolution

Spectral-based [53,54], and spatial-based [9,10] graph convolutional networks are mainly used in traffic forecasting. The spatial-based methods are typically similar to the operation of CNNs, directly aggregating and transforming the features of nodes on the graph. Li et al. proposed diffusion convolution on the graph to capture the spatial dependencies [9]. Compared to standard graph convolutional networks, it can effectively capture long-range spatial dependencies on the adaptive graph. Mathematically, given graph signals $\mathcal{X} \in \mathbb{R}^{N\times F}$, the diffusion process on the graph with a finite K-step truncation can be formulated as follows:

$$Z = \mathcal{X} \star_G W = \sum_{k=0}^{K} AXW_k \tag{2}$$

where Z is the output, $\star_G$ denotes the diffusion convolution operation on the adaptive graph, k is the number of diffusion steps, $A \in \mathbb{R}^{N\times N}$ is the input graph, and $W \in \mathbb{R}^{F\times F}$ denotes the learnable matrix.

3.2.2. Graph attention

General GCNs tend to perform well with static graphs, but their inflexibility becomes a limitation when dealing with dynamic graphs. To address this problem, graph attention network (GAT) [55] is applied to capture dynamic spatial features. By introducing the attention mechanism, GAT can effectively extract nodes' features, especially when the spatial structure changes dynamically [54]. Graph attention on the dynamic graph can be expressed as:

$$\begin{aligned} e_{i,j} &= a([Wh_{i,t} \parallel Wh_{j,t}]),\ j \in N_i \\ a_{i,j} &= \frac{exp(LeakyReLU(e_{i,j}))}{\sum_{k\in N_i} exp(LeakyReLU(e_{i,k}))} \end{aligned} \tag{3}$$

where $e_{i,j} \in \mathbb{R}$ is the similarity coefficient between node i and node j, a is a learnable vector, N_i denotes the neighbor set of node i, and $h_{i,t}$, $h_{j,t} \in \mathbb{R}^F$ denote the attributes of node i and j at time slice t, $W \in \mathbb{R}^{F\times F}$ is a shared matrix for feature transformation. To better reflect the correlation strength between nodes, the similarity coefficients are normalized, where $a_{i,j}$ is the normalized attention coefficient, and $LeakyReLU$ represents the activation function.

4. Methods

4.1. Framework overview

In this section, we present the overall architecture of HSTGNN. As shown in Fig. 2 , HSTGNN consists of three main modules: the temporal decoupling layer (TDL), dual-channel learner (DCL), and hybrid graph learning module (HGL).

The TDL aims to address the challenge of revealing heterogeneous temporal patterns in traffic data. The DCL is designed to further refine the model's ability to learn heterogeneous temporal patterns. This HGL enables the model to capture the dynamic attributes and inherent nature of traffic states and improve prediction accuracy in varying traffic conditions. The details of each module are explained below.

4.2. Decoupling layer

As Fig. 2(a)(c) shows, the TDL leverages discrete wavelet transformation (DWT) to decompose the input traffic sequence into different frequency components, each corresponding to a particular time scale. The low-frequency components represent the long-term trends in traffic conditions, while the high-frequency components denote short-term fluctuations.

In the DWT, the input signal $\mathcal{X} = [X_1, X_2, \ldots, X_T] \in \mathbb{R}^T$ is decomposed into low-frequency (approximation) and high-frequency (detail) components through low- and high-pass filters. At each level, a 1/2 downsampling is applied to reduce the number of data and retain the key information of the series, further obtaining the sub-series, as shown in Fig. 3. Mathematically, at the ith level, the low- and high-frequency sub-series $\mathcal{X}^l(i)$ and $\mathcal{X}^h(i)$ are generated as follows:

$$\begin{aligned} a_n^l(i+1) &= \sum_{k=1}^{K} X_{n+k-1}^l(i) \cdot l_k, \\ a_n^h(i+1) &= \sum_{k=1}^{K} X_{n+k-1}^l(i) \cdot h_k \end{aligned} \tag{4}$$

where $\cdot$ is the dot product, and $X_n^l(i)$ is the nth element of the low-frequency sub-series in the ith level. $a^l(i) = \{a_1^l(i), a_2^l(i), \ldots\}$ and $a^h(i) = \{a_1^h(i), a_2^h(i), \ldots\}$ are intermediate sequences generated by filtering. $l = \{l_1, \ldots, l_K\}$ and $h = \{h_1, \ldots, h_K\}$ are the low-pass and high-pass filters, and $K \ll T$. $X^l(0)$ is set as the original input series. The downsampled sub-series are then expressed as:

$$\begin{aligned} \mathcal{X}^l(i) &= \text{ downsample } \left(a^l(i)\right), \\ \mathcal{X}^h(i) &= \text{ downsample } \left(a^h(i)\right) \end{aligned} \tag{5}$$

At the first decomposition level, the result is denoted as $\{\mathcal{X}^l(1), \mathcal{X}^h(1)\}$. Similarly, the second-level decomposition further divides $\mathcal{X}^l(1)$, resulting in $\{\mathcal{X}^h(1), \mathcal{X}^h(2), \mathcal{X}^l(2)\}$.

To obtain the heterogeneous patterns in the time domain, the Inverse Discrete Wavelet Transform (IDWT) is used to reconstruct the series by upsampling and filtering the low- and high-frequency components. The process of IDWT can be expressed as:

$$\begin{aligned} \tilde{X}_n^l(2) &= \sum_{k=1}^{K} \tilde{a}_{n-k+1}^l(2) \cdot l_k^T, \\ \tilde{X}_n^h(1) &= \sum_{k=1}^{K} \tilde{a}_{n-k+1}^h(1) \cdot h_k^T, \tilde{X}_n^h(2) = \sum_{k=1}^{K} \tilde{a}_{n-k+1}^h(2) \cdot h_k^T \end{aligned} \tag{6}$$

where $\tilde{X}_n(i)$ is the nth element in the reconstructed sub-series, l^T and h^T are the transpose of original filters and $\tilde{a}_n^l(i)$, $\tilde{a}_n^h(i)$ are the upsampled sequences from ith level, expressed as:

$$\begin{aligned} \tilde{a}^l(i) &= \text{upsample}\left(\mathcal{X}^l(i)\right), \\ \tilde{a}^h(i) &= \text{upsample}\left(\mathcal{X}^h(i)\right) \end{aligned} \tag{7}$$

The reconstructed components in the time domain are denoted as $\{\tilde{\mathcal{X}}^h(1), \tilde{\mathcal{X}}^h(2), \tilde{\mathcal{X}}^l(2)\}$. Considering that modeling each high-frequency component individually can be parametrically and computationally burdensome, thus different high-frequency features are mapped into a uniform high-dimensional representation through linear transformation. The trend and event representations $\mathcal{X}_l$, $\mathcal{X}_h \in \mathbb{R}^{T\times N\times D}$ can be obtained by following equations:

$$\begin{aligned} \mathcal{X}_l &= W_l \tilde{\mathcal{X}}^l(2) + b^l, \\ \mathcal{X}_h &= W_{h_1} \tilde{\mathcal{X}}^h(1) + W_{h_2} \tilde{\mathcal{X}}^h(2) + b_h \end{aligned} \tag{8}$$

where $W \in \mathbb{R}^{F\times D}$, $b \in \mathbb{R}^D$ are learnable parameters for linear transformation.

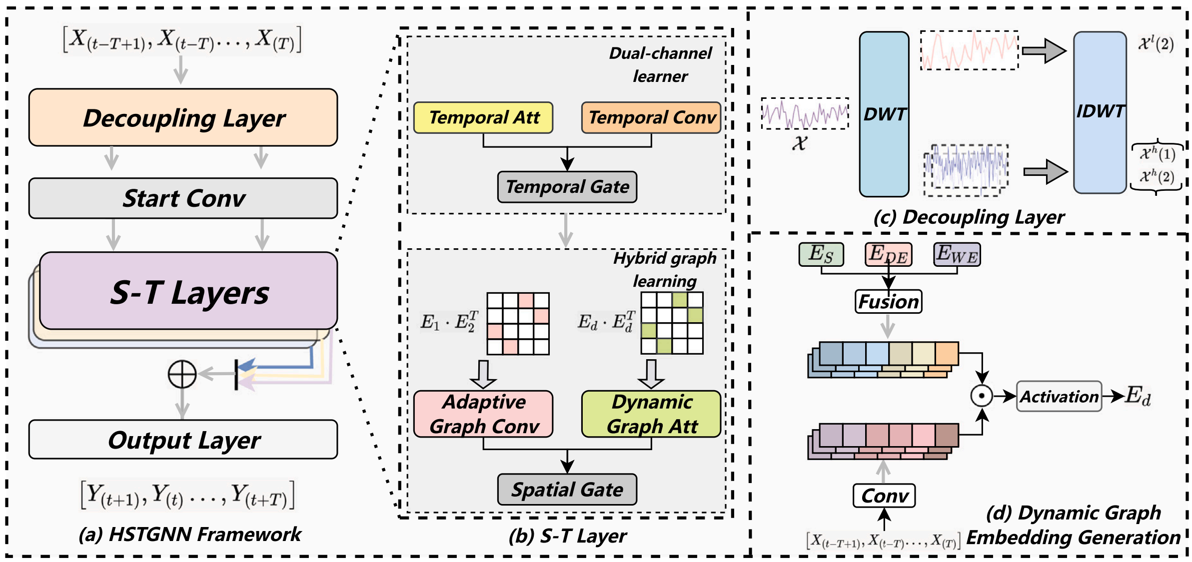

Fig. 2. (a): Framework of HSTGNN. (b) A single spatial–temporal layer of HSTGNN. It consists of two components: dual-channel learner (DCL) and hybrid graph learning module (HGL). (c): The decoupling layer decouples the original series into high- and low-frequency components. The start conv layer uses two 1×1 convolutional networks to map these components into high-dimension space. (d) The dynamic graph embedding module is used to generate a time-varying graph. Fusion: Hadamard product; Conv: 1×1 convolutional layer for linear transformation. Activation: Tanh function.

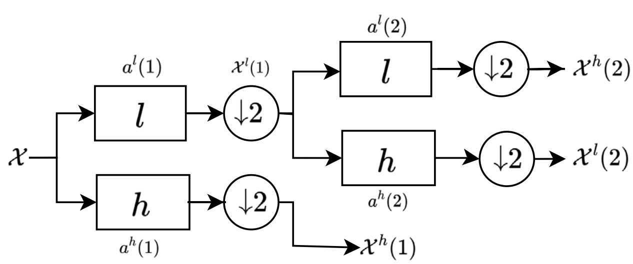

Fig. 3. A schema of two-level discrete wavelet transform. l: Low-pass filter. h: High-pass filter. $\downarrow 2$: 1/2 down-sampling.

4.3. Dual-channel learner

As Fig. 1 shows, the trends represent long-term patterns that are generally stable and persistent over time. These patterns evolve gradually, and distant time slices in the trend still maintain a strong correlation. In contrast, events refer to short-term fluctuations in traffic data. These changes are highly volatile, and only consecutive time slices are closely related. Considering the different properties of these components, we design a dual-channel learner to learn and extract complementary information from heterogeneous temporal features.

4.3.1. Temporal attention

In this channel, we use temporal attention to model the trend representations. It allows the model to interact with the entire historical sequence, capturing the global context and long-term dependencies inherent in the trends. In addition, it can effectively focus on relevant historical information over a long period, which is suitable for modeling stable trends. Given the trend representations $X_l \in \mathbb{R}^{T\times N\times D}$ as input, the formula for the temporal attention on the trends can be expressed as:

$$E = Softmax(V \cdot \sigma((X_l^T W_1)W_2(X_l W_3)^T) + b) \tag{9}$$

where $V \in \mathbb{R}^{N\times N}$, $b \in \mathbb{R}^{T\times N\times N}$, $W_1 \in \mathbb{R}^D$, $W_2 \in \mathbb{R}^{T\times D}$, and $W_3 \in \mathbb{R}^T$ are learnable parameters. Subsequently, all data were normalized by the $softmax$ function. E is the learned score matrix, then we can get the updated output of temporal attention by:

$$Z_{tatt} = X_l E \tag{10}$$

where $Z_{tatt} \in \mathbb{R}^{T\times N\times D}$ denotes the output.

4.3.2. Temporal convolution

As mentioned above, the events represent short-term fluctuations in traffic data, with significant correlations primarily between consecutive time slices. To this end, we utilize dilated causal convolution in one channel [10]. It introduces dilation rates between the kernel elements, effectively expanding the receptive field without increasing the kernel size. This limited receptive field of convolutions can enable the model to capture the transient characteristics of events in traffic data effectively. Given the input $\mathcal{X}_h \in \mathbb{R}^T$, the temporal convolution can be expressed as follows:

$$\mathcal{X}_h \star_\tau f(t) = \sum_{k=0}^{K-1} f(k)\mathcal{X}_h(t - d \times k) \tag{11}$$

where $\star_\tau$ denotes the dilated convolution operation, $f \in \mathbb{R}^K$ is the convolutional kernel, d is dilated factor. The output of the temporal convolution layer can be represented as $Z_{tconv} \in \mathbb{R}^{T\times N\times D}$.

4.3.3. Temporal gated unit

Indeed, the influence of trends and events on traffic conditions can vary at different times. For example, during traffic congestion, the impact of events on traffic states may be greater than the long-term trends. However, during other periods, traffic states may be primarily influenced by trends, with a lower impact from short-term events. To this end, a temporal gated unit is designed to adaptively select the most influential information between trends and events at the current step. The temporal gated unit can be expressed as follows:

$$\begin{aligned} Z_T &= gate \odot Z_{tconv} + (1 - gate) \odot Z_{tatt} \\ gate &= \sigma(Z_{tconv} W_{tc} + Z_{tatt} W_{ta} + b) \end{aligned} \tag{12}$$

where $Z_T \in \mathbb{R}^{T\times N\times D}$ is the output of the temporal module, $W_{ta}, W_{tc} \in \mathbb{R}^{D\times D}$ and $b \in \mathbb{R}^D$ are learnable parameters.

4.4. Hybrid graph learning

Traffic networks inherently exhibit both inherent and dynamic attributes. The inherent attributes reflect the relatively stable structure of the network. In contrast, dynamic attributes represent the features of the traffic network that change over time. Therefore it is essential to design models that can capture these compound spatial correlations. To this end, we design a hybrid graph learning module.

4.4.1. Adaptive graph learning

Since the traffic network structure remains relatively stable from a long-term perspective, we embed node-level attributes to high-dimensional vectors to construct a self-learning graph. It reflects the inherent nature and underlying road network structure and provides a basic framework for the traffic network [10]. Thus we adopt learnable node embedding to obtain adaptive graph A_a by:

$$A_a = softmax(ReLU(E_1 \cdot E_2^T)) \tag{13}$$

where $E_1, E_2 \in \mathbb{R}^{N\times\varphi}$ are learnable node embeddings, N is the number of nodes, φ is the hidden dimension and $ReLU$, $softmax$ functions are used to eliminate weak dependencies and normalized learned matrix. Then we perform diffusion convolution by Eq. (2) on the learned graph and the formula can be expressed as the following equations:

$$Z_{gconv} = \sum_{k=0}^{K} A_a Z_T W_k \tag{14}$$

where $Z_{gconv} \in \mathbb{R}^{T\times N\times D}$ is the output of adaptive graph convolution network.

4.4.2. Dynamic graph learning

To reflect the time-varying properties of the road network structure, we design a dynamic graph generation approach. Specifically, considering rich information is inherently hidden in the traffic data, we encode it into a dynamic graph embedding through linear transformation. Then, we incorporate spatial–temporal embeddings to generate a dynamic spatial–temporal embedding in a data-driven manner. Finally, the embedding is used to derive the dynamic graph structure under specific spatial–temporal context. This approach enables the model to incorporate dynamic spatial semantic information, allowing it to adaptively capture the changing traffic patterns across both space and time. As a result, the model can more effectively learn the underlying dynamics of the traffic network and improve its performance.

Mathematically, let $\mathcal{X} \in \mathbb{R}^{N\times D}$ as the inputs at time slice t, we generate dynamic spatial–temporal graph embedding by Eq. (15):

$$E_d = tanh(W_d \mathcal{X} \odot E_{ST}) \tag{15}$$

where $W_d \in \mathbb{R}^{D\times d}$ is the parameter matrix for linear transformation, $E_{ST} \in \mathbb{R}^{N\times d}$ is learnable spatial–temporal embedding, where $E_{ST} = E_S \odot (E_{WE} \odot E_{DE})$, E_S, E_{DE} and $E_{WE} \in \mathbb{R}^{N\times d}$ are learnable spatial, daily and weekly embedding matrices at time slice t. Finally, similar to obtaining an adaptive graph like Eq. (13), the spatial–temporal dynamic graph can be calculated by:

$$A_d = softmax(ReLU(E_d \cdot E_d^T)) \tag{16}$$

where $A_d \in \mathbb{R}^{N\times N}$ denotes the learned dynamic spatial–temporal graph.

4.4.3. Spatial gated unit

After obtaining the dynamic graph, our HSTGNN aggregates the dynamic spatial features by Eq. (3), and in this paper, we employ a multi-head attention mechanism in GAT. The multi-head attention representation of node i can be calculated by the following equation:

$$h'_{i,t} = \sigma(\frac{1}{K}\sum_{k=1}^{K}\sum_{i\in N_i} a_{i,j} W^k h_{j,t}) \tag{17}$$

where $h'_{i,t} \in \mathbb{R}^D$ denotes the learned node features at time slices t, and K is the number of heads. The final output of GAT can be presented by $Z_{gatt} \in \mathbb{R}^{T\times N\times D}$. We obtain the output of the hybrid graph learning module by a spatial gated unit:

$$\begin{aligned} Z_S &= gate \odot Z_{gconv} + (1-gate)\odot Z_{gatt} \\ gate &= \sigma(Z_{gconv} W_{gc} + Z_{gatt} W_{ga} + b) \end{aligned} \tag{18}$$

where $Z_S \in \mathbb{R}^{T\times N\times D}$ denotes the output of our hybrid graph learning module, W and b are learnable parameters.

4.5. Output layer

In this work, we employ a stacking structure and fuse multi-layer information to capture spatial and temporal features. We apply the residual connections to enhance the model's convergence speed, facilitating the retention of crucial information throughout the learning process. The output of spatial-temporal layers can be calculated by:

$$Z_{out} = Z_S + Z_T \tag{19}$$

where Z_{out} is the output of HGL module. Rather than relying on a recursive manner to generate prediction results, we design an output layer to achieve multi-step prediction. The formula can be expressed as follows:

$$\hat{Y} = FC_{out}(Z_{out}) \tag{20}$$

where $\hat{Y}$ is the predicted results of HSTGNN, FC_{out} denotes the fully connected network.

Loss function: Mean Absolute Error(MAE) is used as the loss function to optimize our model, which can be expressed as:

$$Loss = \frac{1}{T\times N\times D}\sum_{t=1}^{T}\sum_{n=1}^{N}\sum_{d=1}^{D}\left|\hat{Y}_{t,n,d} - Y_{t,n,d}\right| \tag{21}$$

where $Loss$ denotes the MAE loss function, $\hat{Y}_{t,:,:}$ and $Y_{t,:,:}$ are the predicted values and the ground truth of all nodes at time slice t.

5. Experiments

In this section, we conduct extensive experiments on six real-world datasets to answer the following six research questions:

- **RQ1:** How does the performance of HSTGNN compare to state-of-the-art models?
- **RQ2:** What is the impact of each component on the overall performance of the model?
- **RQ3:** How does the efficiency of HSTGNN compare to baseline models?
- **RQ4:** How do hyperparameters influence the performance of HSTGNN?
- **RQ5:** How robust is HSTGNN under various conditions?
- **RQ6:** How the decoupling layer and hybrid graph work in the model?
- **RQ7:** How does HSTGNN perform in real-world scenarios?

5.1. Datasets

To evaluate the performance of the proposed model, we conduct experiments on six real-world traffic datasets, including METR-LA, PEMS-BAY, PEMS03, PEMS04, PEMS08, and Urban-core datasets. The first five datasets represent coarse-grained traffic scenarios and Urban-core represents fine-grained scenarios.

METR-LA is a highway traffic flow dataset, collected from 207 loop detectors in Los Angeles City. The dataset contains three months of data aggregated in the frequency of 5 min.

PEMS-BAY is a highway speed dataset collected by the California Transportation Agencies (CalTrans) Performance Measurement System

Table 1
The statistics of the datasets.

Dataset	Nodes	Interval	Samples	Data type
METR-LA	207	5 min	34 272	F
PEMS-BAY	325	5 min	52 116	S
PEMS03	358	5 min	26 208	S & F & O
PEMS04	307	5 min	16 992	S & F & O
PEMS08	170	5 min	17 856	S & F & O
Urban-core	304	5 min	8640	S

(PEMS). It includes six months of data, sampled at 5-minute intervals, from 325 loop detectors located in the BAY Area.

PEMS03, PEMS04, and PEMS08 datasets are multi-feature datasets, including three traffic measures: traffic speed, flow, and occupancy. The PEMS03 dataset covers traffic information from California between September 1, 2018, and November 30, 2018. PEMS04 covers the San Francisco BAY Area from January 1, 2018, to February 28, 2018, and includes data from 307 loop detectors. PEMS08 covers the San Bernardino area from July 1, 2016, to August 31, 2016, with data collected from 170 loop detectors.

To further validate the performance in different scenarios, we use the Urban-core [45], an urban dataset comprising taxi speed data collected at 5-minute intervals from 304 sensors located on major roads in Seoul.

Table 1 shows more details about used datasets, where 'S' denotes traffic speed, 'F' denotes flow, and 'O' stands for occupancy.

5.2. Experiment settings

For metrics, this work adopts three commonly used performance metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). MAE denotes the average of the absolute values of the predicted and true values. RMSE denotes the rooted average squared difference between the predicted values and the ground truth. MAPE is the mean absolute percentage error. The metrics are defined as follows:

- $MAE = \frac{1}{M}\sum_{i=1}^{M}\left|\hat{Y} - Y\right|$
- $RMSE = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(\hat{Y} - Y)^2}$
- $MAPE = \frac{100}{M}\sum_{i=1}^{M}\left|\frac{\hat{Y}-Y}{Y}\right|$

where Y denotes the true values, $\hat{Y}$ denotes the predicted values, and M represents the number of values to be predicted.

In the training process, we set the batch size to 64. The initial learning rate is set to 0.001. For the experiments of HSTGNN, in the temporal module, the kernel size in the temporal convolutional network is set to 2. In the spatial module, we set the heads of GAT to 2, and the hidden dimension of spatial–temporal dynamic graph embeddings to 6. The level of DWT is set to 2. For the datasets, the division ratios of training, validation, and test sets for all datasets are 0.7, 0.1, and 0.2 respectively. Besides, we apply different discrete wavelets for different datasets according to experimental results: Haar wavelet for PEMS-BAY and Urban-core, Symlets wavelet for PEMS03, Daubechies wavelet for PEMS04 and METR-LA, Coiflets wavelet for PEMS08. The model was trained in a Pytorch environment with one NVIDIA V100 with 32 GB memory (GPU) and Intel(R) Xeon(R) Platinum 8358P CPU @ 2.60 GHz (CPU).

5.3. Baseline models

To verify the validity of our method, we compare HSTGNN with the classical methods and state-of-the-art methods.

- **HA** [24]: HA is a classic statistical method that uses historical average data to predict future data.
- **STGCN** [40]: STGCN is a purely convolutional operation-based model that combines graph convolution and 1D temporal convolution.
- **DCRNN** [9]: DCRNN models the interaction between road nodes as a diffusion process and the diffusion convolution operation is proposed to capture the spatial dependencies.
- **GWNET** [10]: GWNET captures hidden spatial–temporal dependencies by combining adaptive graph structures and 1-D dilated convolution.
- **ASTGCN** [47]: ASTGCN utilizes temporal attention and spatial convolution to extract dynamic spatial–temporal correlations.
- **AGCRN** [56]: AGCRN captures spatial dependencies by adaptive graph learning and GRU is used to extract temporal features.
- **Ada-STNet** [57]: Ada-STNet uses a graph structure learning component to obtain an optimal graph adjacency matrix from both macro and micro perspectives.
- **PGCN** [45]: PGCN constructs progressive adjacency matrices by learning trend similarities among graph nodes and the model is combined with the dilated causal convolution and gated activation unit to extract temporal features.
- **STPGNN** [17]: STPGNN is a graph neural network-based traffic prediction method that improves prediction performance by identifying the pivotal nodes in the road network as well as designing pivotal graph convolution modules to capture the complex spatial–temporal dependencies.

5.4. Performance comparison (RQ1)

In this section, we conduct two sets of experiments to evaluate the model's performance: regular traffic forecasting and long-term traffic forecasting. Regular traffic forecasting focuses on standard datasets commonly used in research to assess a model's core performance. In comparison, long-term traffic forecasting evaluates a model's ability to make predictions over extended periods, such as 60-time steps into the future. This capability is critical for strategic planning tasks, including route optimization, infrastructure development, and traffic management in urban environments. The results and analysis are as follows:

5.4.1. Results and analysis on regular traffic forecasting

Table 2 compares the performance of various models for 15-, 30-, and 60-minute predictions across six datasets. HA, a classical statistical method, consistently lags behind deep learning methods. STGCN, which integrates GCN and 1-D convolutions, improves accuracy but suffers from error accumulation due to its single-step forecasting strategy. DCRNN performs well on METR-LA and PEMS-BAY but falls short on PEMS03, PEMS04, and PEMS08. GWNET achieves competitive results on PEMS03 but is slightly less effective on other datasets. ASTGCN, considering temporal periodicity, performs poorly due to its predefined graph. Ada-STNet generates graph structures from both macro and micro perspectives, outperforming previous methods on both single and multi-feature datasets. PGCN slightly outperforms STPGNN on METR-LA and PEMS-BAY but underperforms on PEMS03, PEMS04, PEMS08, and Urban-core.

From the table, we can observe that HSTGNN achieves the best performance in at least one metric across all datasets for the 15-, 30-, and 60-min predictions. Additionally, we can have the following findings:

- The statistic method, such as HA, performs inferiorly compared to spatial–temporal neural networks. This shows its inability to capture the complex, non-linear dependencies.
- Models considering dynamic spatial correlations (e.g., HSTGNN, PGCN, and STPGNN) typically perform better than static approaches (e.g., STGCN, DCRNN, and GWNET) because they can adapt to the changing traffic patterns and capture complex spatial dependencies.

Table 2
Performance comparison on regular traffic forecasting. **Bold:** Best.

Dataset	Method	@Horizon 3			@Horizon 6			@Horizon 12		
		MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
METR-LA	HA	4.15	7.77	12.90	4.15	7.77	12.90	4.15	7.77	12.90
	STGCN	2.88	5.74	7.62	3.47	7.24	9.57	4.59	9.40	12.70
	DCRNN	2.70	5.25	6.94	3.10	6.36	8.48	3.58	7.60	10.39
	GWNET	2.69	5.14	6.97	3.07	6.17	8.46	3.52	7.28	10.01
	ASTGCN	4.22	7.92	10.21	5.07	9.35	12.45	6.33	10.97	15.36
	AGCRN	2.85	5.52	7.59	3.23	6.51	9.02	3.61	7.47	10.51
	Ada-STNet	2.65	5.08	6.78	3.03	6.13	8.24	3.48	7.25	9.95
	PGCN	2.69	5.14	6.86	3.08	6.19	8.29	3.54	7.35	9.89
	STPGNN	2.83	5.47	7.42	3.24	6.52	8.91	3.74	7.63	10.60
	HSTGNN	**2.64**	**5.06**	**6.77**	**3.00**	**6.07**	**8.19**	**3.43**	**7.14**	**9.80**
PEMS-BAY	HA	2.88	5.59	6.76	2.88	5.59	6.76	2.88	5.59	6.76
	STGCN	1.36	2.96	2.90	1.81	4.27	4.17	2.49	5.69	5.79
	DCRNN	1.31	2.78	2.74	1.65	3.78	3.70	1.96	4.61	4.64
	GWNET	1.39	3.00	2.91	1.83	4.22	4.14	2.35	5.42	5.80
	ASTGCN	1.70	3.35	3.85	2.02	4.10	4.74	2.37	4.75	5.67
	AGCRN	1.37	2.89	2.95	1.70	3.88	3.90	1.97	4.61	4.70
	Ada-STNet	1.31	2.76	2.77	1.64	3.71	3.75	1.92	4.43	4.60
	PGCN	1.31	**2.74**	2.73	**1.63**	**3.68**	**3.66**	1.93	4.43	4.62
	STPGNN	1.34	2.84	2.79	1.69	3.80	3.78	2.04	4.62	4.77
	HSTGNN	**1.30**	2.76	**2.72**	**1.63**	3.70	3.68	**1.91**	**4.42**	**4.56**
PEMS03	HA	24.91	45.99	25.80	24.91	45.99	25.80	24.91	45.99	25.80
	STGCN	14.97	26.68	16.41	16.38	28.78	18.14	18.76	32.43	18.42
	DCRNN	14.67	25.40	14.26	16.32	28.07	15.46	19.10	32.41	17.99
	GWNET	13.45	23.17	**13.47**	14.52	25.23	14.29	16.17	27.87	**15.10**
	ASTGCN	15.75	26.33	14.97	17.98	30.08	16.69	22.72	37.44	21.04
	AGCRN	14.01	24.75	13.84	15.23	26.75	14.34	16.95	29.23	16.28
	Ada-STNet	13.47	23.14	13.85	14.53	25.31	**14.26**	16.11	27.85	15.60
	PGCN	13.82	23.84	14.00	14.82	25.48	15.15	16.76	28.23	16.87
	STPGNN	13.42	23.15	14.16	14.39	24.55	15.47	16.14	27.48	17.26
	HSTGNN	**13.39**	**23.12**	14.20	**14.36**	**24.37**	15.51	**16.08**	**27.35**	17.33
PEMS04	HA	31.58	52.39	33.78	31.58	52.39	33.78	31.58	52.39	33.78
	STGCN	18.78	29.93	13.33	19.70	31.38	13.94	21.55	33.88	13.92
	DCRNN	18.79	29.65	13.16	20.05	31.51	14.08	20.19	**31.64**	14.24
	GWNET	18.17	28.98	13.14	19.18	30.54	13.72	21.04	33.14	15.12
	ASTGCN	19.17	30.61	12.92	20.32	32.39	13.90	23.31	36.73	16.06
	AGCRN	19.11	31.40	12.65	19.84	32.71	13.10	21.46	35.38	**14.08**
	Ada-STNet	17.81	28.68	**12.38**	18.66	30.02	13.09	20.06	32.01	14.29
	PGCN	18.02	28.85	12.96	19.02	30.35	13.68	20.76	32.70	14.96
	STPGNN	17.70	28.56	**12.38**	18.50	29.88	**12.83**	20.12	32.12	14.14
	HSTGNN	**17.67**	**28.55**	12.77	**18.43**	**29.73**	13.51	**19.90**	31.75	14.38
PEMS08	HA	34.86	59.24	27.88	34.86	59.24	27.88	34.86	59.24	27.88
	STGCN	14.55	22.72	9.38	15.62	24.55	9.84	17.86	27.56	11.14
	DCRNN	14.21	22.19	9.19	15.18	24.01	9.79	16.97	26.90	10.94
	GWNET	13.68	21.61	8.80	14.65	23.43	9.39	16.42	26.25	10.75
	ASTGCN	16.37	25.31	9.98	18.33	28.28	10.97	22.44	34.11	13.24
	AGCRN	14.78	23.27	9.49	15.77	25.01	10.17	17.52	27.69	11.25
	Ada-STNet	13.48	21.51	8.78	14.33	23.24	9.36	15.69	25.51	**10.25**
	PGCN	13.44	21.53	8.84	14.25	23.12	9.42	15.54	**25.31**	10.28
	STPGNN	13.26	21.47	8.77	14.14	23.39	**9.16**	15.70	25.82	10.27
	HSTGNN	**13.11**	**21.36**	**8.51**	**13.94**	**23.09**	**9.16**	**15.47**	25.50	10.40
Urban-core	HA	3.05	4.68	11.77	3.47	5.16	13.69	4.03	5.83	16.25
	STGCN	2.44	3.74	9.54	2.66	3.99	10.63	2.88	4.26	11.75
	DCRNN	2.43	3.73	9.53	2.65	3.99	10.68	2.84	4.23	11.62
	GWNET	2.40	3.70	9.31	2.60	3.93	10.38	2.75	4.12	11.17
	ASTGCN	2.53	3.93	9.91	2.72	4.19	10.85	2.90	4.37	11.93
	AGCRN	2.48	3.81	9.76	2.64	4.00	10.57	2.84	4.25	11.52
	Ada-STNet	2.40	3.70	9.36	2.61	3.94	10.43	2.78	4.15	11.28
	PGCN	2.41	3.72	9.37	2.63	3.98	10.46	2.83	4.21	11.44
	STPGNN	2.40	3.71	**9.27**	2.60	3.94	**10.30**	2.76	4.11	11.13
	HSTGNN	**2.39**	**3.69**	9.29	**2.58**	**3.92**	10.32	**2.73**	**4.10**	**11.06**

- HSTGNN achieves comprehensively optimal performance on both single-feature and multi-feature datasets, demonstrating its effectiveness. This can be attributed to the ability to capture complex and time-varying spatial dependencies through the hybrid graph learning approach efficiently. In addition, HSTGNN can model heterogeneous temporal patterns, leading to improved prediction accuracy and generalization across diverse datasets.

5.4.2. *Results and analysis on long-term traffic forecasting*

As mentioned above, the decoupling layer and dual-channel learning module are effective in capturing heterogeneous temporal patterns, making them suitable for long-term forecasting. Specifically, these mechanisms enable the model to leverage complementary information embedded in diverse temporal patterns, thereby better handling the complexity and uncertainty inherent in long-term traffic prediction tasks. In Table 3, we further conduct experiments on three datasets to

Table 3
Performance comparison on long-term traffic forecasting. **Bold:** Best.

Dataset	Method	@Horizon 36			@Horizon 48			@Horizon 60			Average		
		MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE	MAE	RMSE	MAPE (%)
PEMS03	STGCN	22.59	37.87	23.33	23.92	40.31	26.07	25.13	42.55	29.70	21.33	36.05	22.35
	DCRNN	23.24	40.57	23.76	25.00	44.09	25.88	26.39	46.72	28.00	21.63	37.59	22.34
	GWNET	22.79	38.66	24.23	23.98	39.17	25.78	25.64	42.03	28.34	22.01	36.14	21.79
	ASTGCN	24.67	39.77	24.79	26.49	42.61	27.60	30.21	47.47	33.56	23.22	38.05	23.35
	AGCRN	22.23	37.08	21.92	23.81	39.44	24.44	25.22	41.92	27.76	21.13	35.71	21.03
	Ada-STNet	20.97	35.62	20.86	22.44	38.74	22.20	23.89	40.83	23.72	20.09	34.15	19.71
	PGCN	20.98	34.57	22.59	22.55	37.29	23.93	24.59	40.63	26.16	20.17	33.56	21.35
	STPGNN	20.49	34.34	23.05	22.63	37.71	26.43	25.36	42.58	29.89	19.62	32.91	21.11
	HSTGNN	**20.28**	**32.40**	**20.31**	**22.20**	**36.49**	**21.21**	**23.76**	**39.80**	**22.82**	**19.15**	**31.83**	**19.38**
PEMS08	STGCN	21.32	33.03	14.70	23.05	35.35	15.69	24.88	37.88	17.19	20.47	31.64	14.05
	DCRNN	20.48	31.73	13.73	21.38	33.03	14.56	23.12	35.22	16.35	20.02	30.93	13.58
	GWNET	21.27	35.06	13.75	21.54	35.76	13.96	22.01	36.38	14.45	21.00	34.63	13.63
	ASTGCN	29.32	42.91	19.76	31.63	46.27	21.19	36.10	52.47	25.35	27.21	40.87	18.55
	AGCRN	21.29	33.84	14.50	22.53	35.78	15.52	24.73	38.79	17.15	20.57	32.75	13.93
	Ada-STNet	19.52	31.30	**12.77**	20.81	32.92	13.89	22.30	34.71	14.97	19.05	30.36	12.56
	PGCN	19.82	31.99	13.42	20.86	33.60	14.17	22.30	35.59	15.66	19.20	30.74	13.42
	STPGNN	19.56	31.79	13.03	20.46	33.29	13.93	21.92	35.21	15.17	18.98	30.70	12.80
	HSTGNN	**18.09**	**29.47**	12.81	**18.53**	**30.33**	**13.34**	**19.37**	**31.29**	**14.55**	**17.30**	**28.20**	**12.50**
Urban-core	STGCN	2.92	4.31	12.06	2.94	4.34	12.18	3.04	4.47	12.71	2.87	4.25	11.86
	DCRNN	2.88	4.30	11.80	2.90	4.31	11.90	2.95	4.37	12.16	2.83	4.23	11.55
	GWNET	2.90	4.33	11.79	2.91	4.34	11.84	2.97	4.41	12.19	2.85	4.27	11.61
	ASTGCN	3.27	4.72	13.65	3.36	4.83	14.12	3.58	5.06	15.34	3.19	4.61	13.31
	AGCRN	3.02	4.44	12.39	3.05	4.47	12.47	3.10	4.51	12.73	3.07	4.33	12.09
	Ada-STNet	2.97	4.37	12.28	2.99	4.39	12.37	3.06	4.47	12.68	2.90	4.28	11.92
	PGCN	2.85	4.23	11.79	2.87	4.25	11.80	2.97	4.37	12.21	2.84	4.21	11.69
	STPGNN	2.83	4.19	11.77	2.86	4.20	11.81	2.98	4.38	12.24	2.86	4.22	11.71
	HSTGNN	**2.79**	**4.13**	**11.63**	**2.83**	**4.15**	**11.65**	**2.95**	**4.33**	**12.07**	**2.80**	**4.17**	**11.53**

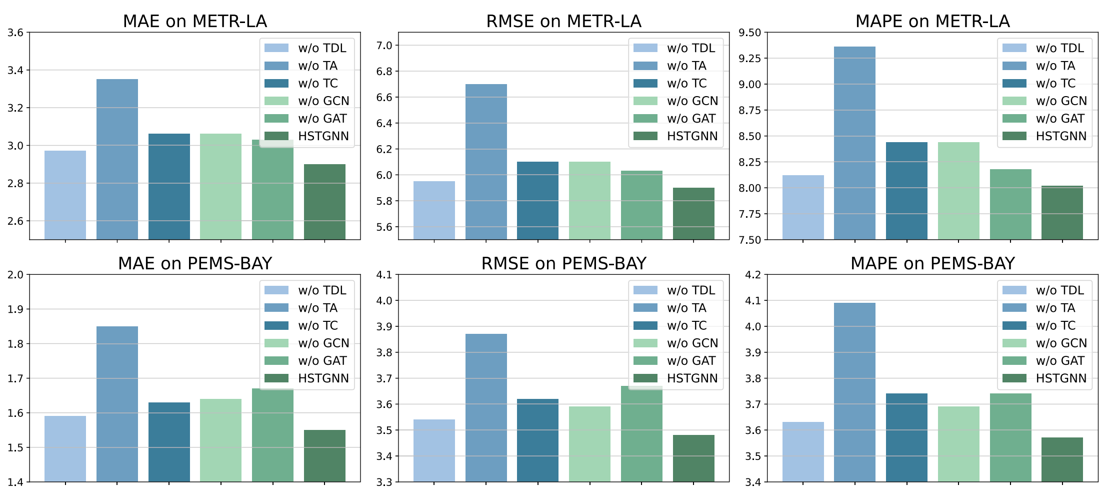

Fig. 4. Ablation study on METR-LA and PEMS-BAY datasets. w/o TDL: remove decoupling layer. w/o TA: remove temporal attention. w/o TC: remove temporal convolutional network. w/o GCN: remove GCN. w/o GAT: remove GAT.

predict traffic states over 60 future horizons (5 h). This experiment aims to highlight the suitability and generalization capability of HSTGNN for long-term traffic forecasting. From the results, we observe that HSTGNN consistently outperforms other models, achieving the best prediction performance across all datasets. This finding demonstrates the effectiveness and generalization in long-term prediction tasks. Notably, when transitioning from short-term to long-term forecasting, all models exhibited a significant increase in prediction error. For example, in terms of MAE on the PEMS08 dataset, PGCN's MAE increased by 35.02%, STPGNN's by 33.94%, whereas HSTGNN's error only increased by 23.31%. Similarly, on PEMS03, the error growth rates were 35.01%, 36.16%, and 33.36% respectively, and on Urban-core, they were 9.23%, 12.16%, and 12.45%. This bias can be partially attributed to dataset-specific characteristics, as each dataset exhibits distinct data distributions. Additionally, different methods vary in their feature extraction capabilities. While models with simple architectures may excel in short-term forecasting, they often struggle to capture long-term dependencies. We argue that treating heterogeneous components as a single entity leads to oversimplified representations, causing models to misinterpret transient anomalies as persistent patterns and fail to capture essential long-term dependencies.

To further address the potential biases introduced by different data distributions and model architectures, our future work will focus on developing adaptive mechanisms capable of generalizing across diverse datasets while maintaining robustness to heterogeneous temporal and spatial patterns.

5.5. Ablation study (RQ2)

To explore the validity of different components in HSTGNN, we conduct an ablation study on METR-LA and PEMS-BAY datasets. Specifically, we compare HSTGNN with the following variants, including:

Table 4
Ablation study of hybrid graph processing architectures on PEMS04 and PEMS08 datasets.

Methods	PEMS04			PEMS08		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
GAT-GCN	22.17	35.00	15.29	15.71	25.61	10.26
GCN-GCN	18.89	30.38	**13.34**	14.09	**23.16**	9.67
GAT-GAT	22.44	35.24	15.49	16.97	26.90	10.95
GCN-GAT (Ours)	**18.46**	**29.72**	13.37	**14.03**	23.36	**9.34**

- **w/o TDL:** HSTGNN removes the decoupling layer.
- **w/o TA:** HSTGNN removes the temporal attention and no longer considers the trend representations in the traffic data.
- **w/o TC:** HSTGNN removes the temporal convolutional network and no longer considers the event representations in the traffic data.
- **w/o GCN:** HSTGNN removes the GCN and only adopts the dynamic graph to model time-varying spatial changes.
- **w/o GAT:** HSTGNN removes the GAT and only adopts the adaptive graph to model spatial features.

Fig. 4 shows the performance comparison between these variants and HSTGNN. It is observed that the original HSTGNN outperforms all the variants. In addition, we can obtain some insightful information: (1) When we remove the decoupling layer, the performance of the model decreases, indicating the importance of explicitly modeling the heterogeneous temporal patterns. (2) The model's performance significantly decreases when solely relying on event information within the time series, as trends play a more crucial role in long-term forecasting. (3) The 'w/o GAT' results show the significance of modeling dynamic spatial dependencies. Without the inclusion of the dynamic graph, the model fails to capture crucial spatial context. In Table 4, we further compare different GNN combinations to validate the rationality of the proposed hybrid graph learning module. Specifically, we explore the performance of different GCN and GAT combinations on the PEMS04 and PEMS08 datasets., the GCN-GAT combination adopted in HSTGNN performs best on both datasets, suggesting its rationality and effectiveness.

5.6. Computation efficiency (RQ3)

In Fig. 5, we compare HSTGNN with state-of-the-art methods on the PEMS08 dataset to demonstrate the efficiency, covering regular and long-term traffic forecasting tasks. From the figure, we can have the following observations: (1) The baseline methods exhibit a noticeable increase in prediction errors, training time, and memory usage as the prediction horizon extends. (2) STGCN demonstrates relatively short training time and low memory usage due to its simple architecture. However, this simplicity comes at the cost of reduced prediction accuracy, particularly in more complex, long-term forecasting scenarios. (3) Our HSTGNN maintains moderate training time and memory usage compared to other models. Additionally, the well-designed architecture leads to enhanced prediction performance, especially in long-term forecasting tasks.

5.7. Parameter sensitivity analysis (RQ4)

To further explore the impact of various parameters, we perform sensitivity analysis for HSTGNN. Specifically, we explored various values for each hyperparameter within predefined search spaces: [1, 2, 4, 8] for the number of heads in GAT, [6, 12, 24, 32] for the hidden dimension in the dynamic graph embedding, [Haar, Daubechies, Symlets, Coiflets, Biorthogonal] for the wavelet and [1, 2, 3] for the level in the decoupling layer. This comprehensive analysis allowed us to evaluate the impact of different configurations on the performance of our HSTGNN. The results are shown in Fig. 6.

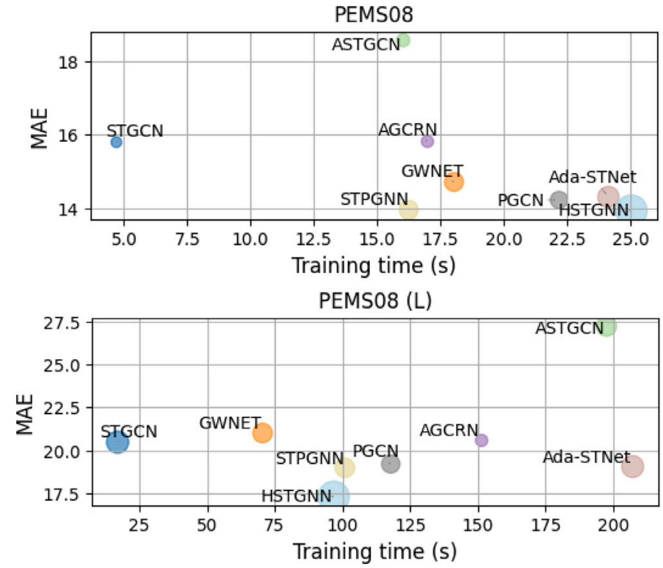

Fig. 5. Computation efficiency comparison in regular (PEMS08) and long-term (PEMS08 (L)) forecasting tasks. MAE denotes the average error. A larger scatter plot indicates higher memory usage. We have not included DCRNN in the figure because its training time far exceeds the rest of the models.

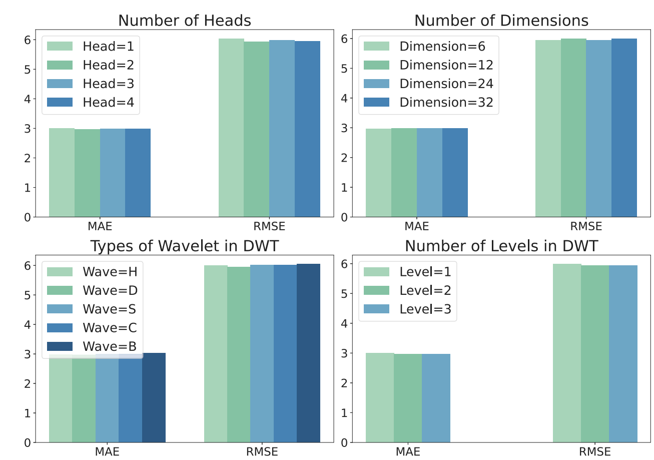

Fig. 6. Experimental results of the hyper-parameter study on the METR-LA dataset. H: Haar, D: Daubechies, S: Symlets, C: Coiflets, B: Biorthogonal.

From the figure, we have the following observations: (1) Increasing the number of heads in GAT allows for exploring richer spatial information. However, the performance improvement can decrease progressively and eventually stabilize. (2) The hidden dimension set to 6 can achieve the lowest errors. Larger dimensions increase the model's capacity of representation, allowing it to capture more complex features. There is no significant improvement after increasing the value, suggesting that the model has captured enough information. (3) The choice of wavelet affects the model's ability to capture data patterns because different wavelet functions have different decoupling properties. For instance, Daubechies wavelets offer smoother transitions and are more effective at capturing gradual variations in traffic patterns over time. In contrast, Haar wavelets may be better suited for capturing sharp transitions or discontinuities in the data. (4) The DWT level influences decomposition granularity, affecting the model's performance. A higher DWT level allows for the finer decomposition of the data, enabling the model to capture more detailed short-term fluctuations. In comparison, a lower level may focus more on long-term trends.

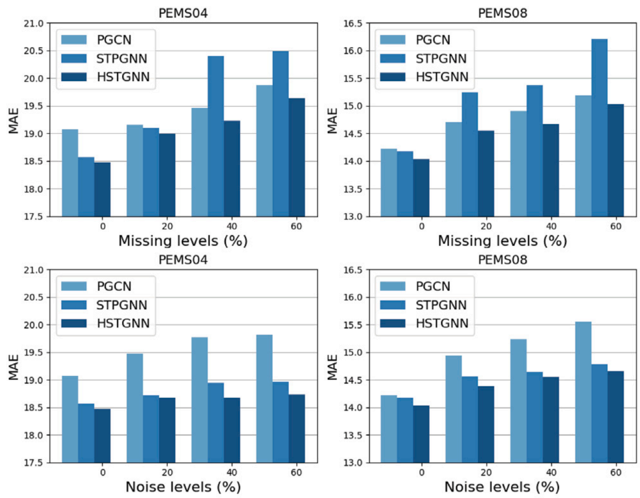

Fig. 7. Performance comparison of HSTGNN, PGCN, and STPGNN on the PEMS04 and PEMS08 datasets under different noisy and missing data levels.

5.8. Robustness testing (RQ5)

In real-world applications, the collected data is often affected by various internal or external factors, leading to missing or inaccurate data. Thus, the model should handle these challenges effectively to maintain stable and reliable performance. To assess the model's robustness in real-world scenarios, we design and conduct two types of experiments:

- **Data missing**: We simulate real-world data loss by removing 20%, 40%, and 60% of the training data, representing mild, moderate, and severe levels of data missing.
- **Data noise**: To model real-world uncertainties, we introduce random Gaussian noise with varying intensities (10%, 20%, and 30%) into the training data, with a mean of 10 and a standard deviation of 500, representing mild, moderate, and severe levels of data pollutions, to evaluate the model's stability against noisy inputs.

As shown in Fig. 7, PGCN showed a noticeable increase in MAE as noise levels increased. Compared to PGCN, the performance of STPGNN is slightly degraded but still better than PGCN. In terms of data-missing, PGCN outperforms STPGNN. At lower missing ratios (20%), PGCN and STPGNN perform comparably. However, when facing higher missing ratios (>40%), STPGNN's error increases significantly. HSTGNN demonstrates exceptional robustness across different types and levels of anomalies, particularly under high missing ratios and noise levels. This performance can be attributed to its well-designed architecture. The TDL can isolate temporal dependencies from noise, enabling the DCL to focus on useful information. The results indicate that HSTGNN is suited for real-world complex traffic scenarios involving various data anomalies, offering better practicality.

5.9. Effect of decoupling layer and hybrid graph (RQ6)

As shown in Fig. 8, we conducted experiments on PEMS04 and PEMS08 datasets to explore the interpretability of the trends and events features learned through the decoupling layer. Specifically, $\mathcal{X}_l$ and $\mathcal{X}_h$ represent the trend and event features, respectively, and three types of experiments were conducted to evaluate their contributions:

- Trends and events: both $\mathcal{X}_l$ and $\mathcal{X}_h$ are used as input to the dual-channel learner, representing the complete decoupling layer design.
- Trends only: Only the low-frequency trends are used as input to the dual-channel learner.
- Events only: Only the high-frequency events are used as input to the dual-channel learner.

In Fig. 8(a) and (d), we compare HTSGNN with STPGNN and visualize the prediction curves on PEMS04 (node 197) and PEMS08 (node 74). The results demonstrate that HSTGNN aligns well with ground truth and consistently outperforms STPGNN in capturing both long-term trends and short-term fluctuations in traffic flow. In terms of short-term fluctuations, the zoomed-in regions reveal that HSTGNN is significantly better at modeling rapid changes in traffic flow while STPGNN often exhibits notable deviations from the ground truth. We argue that the decoupling layer effectively separates heterogeneous temporal patterns, ensuring that each temporal component is learned independently, reducing the interference between global and local patterns.

From Fig. 8(b) and (e), we can observe that when HSTGNN is provided with only trend features as input, the model can gradually adjust its predictions in the direction of the correct trend. However, the lack of high-frequency events restricts the model's ability to respond dynamically during periods of significant fluctuation, confining the variations to a narrow range and failing to capture abrupt changes effectively. When HSTGNN is provided with only event features as input, the model hardly fits long-term trends, leading to excessive fluctuations, as shown in Fig. 8(c) and (f). The absence of trends features reduces the model's sensitivity to forecast directions, resulting in overly influenced by localized variations.

In short, Fig. 8 demonstrates that HSTGNN effectively addresses the heterogeneity in traffic sequences by decoupling trends and events, enabling accurate modeling of both abrupt changes and gradual variations. In contrast, models like STPGNN do not account for the heterogeneous temporal patterns of traffic sequences, limiting their ability to adapt to dynamic fluctuations and leading to less accurate predictions in both sharp and smooth trends. This emphasizes the importance of our decoupling layer and incorporating temporal heterogeneity for robust traffic forecasting.

To intuitively illustrate the impact of the proposed hybrid graph structure learning component, we conduct visual experiments in Fig. 9. For clarity, we only use the first 30 nodes. Fig. 9(a) shows the distribution of the first 30 nodes while Fig. 9(b) shows the distribution of sampled sensor 1, 2 and 27, 28. Fig. 9(c) and (d) show the heatmaps of the all-hours predefined adjacency matrix and adaptive matrix. The predefined matrix is calculated by geographic distance.

From Fig. 9(a)(b)(d) we can observe that although sensor 1 and sensor 2 are relatively close and present a strong relation in the predefined matrix, they exhibit different traffic patterns within one day in Fig. 9(h). In the adaptive adjacency matrix, they are also not strongly connected. That is, a predefined matrix hardly reflects real-world road conditions. Indeed, sensors 1 and 2 are in opposite directions on adjacent roads, shown in Fig. 9(b). Despite the proximity of Euclidean distance, the traffic conditions are different. The same applies to nodes 27 and 28 as they are in opposite directions of the road and exhibit different traffic patterns. Furthermore, we observe that node 1 and node 27 have similar traffic patterns, while node 2 and node 28 also have similar patterns. This is because they are located in the same direction on one road. Fig. 9(e)(f) denotes the dynamic adjacency matrices during peak and off-peak traffic hours on the METR-LA dataset. During peak hours, certain nodes become more strongly connected, reflecting the higher traffic flow between these areas, while off-peak hours reveal weaker connections, as traffic decreases. The variation in heat maps also reflects the regularity and time-varying nature of the traffic state. Peak hours tend to show patterns of congestion or dense traffic flows, while off-peak periods show relatively smooth traffic flows. This dynamic change suggests that the spatial and temporal relationships of the road network are constantly changing over time, and a fixed adjacency matrix cannot capture these changes. In summary, from both

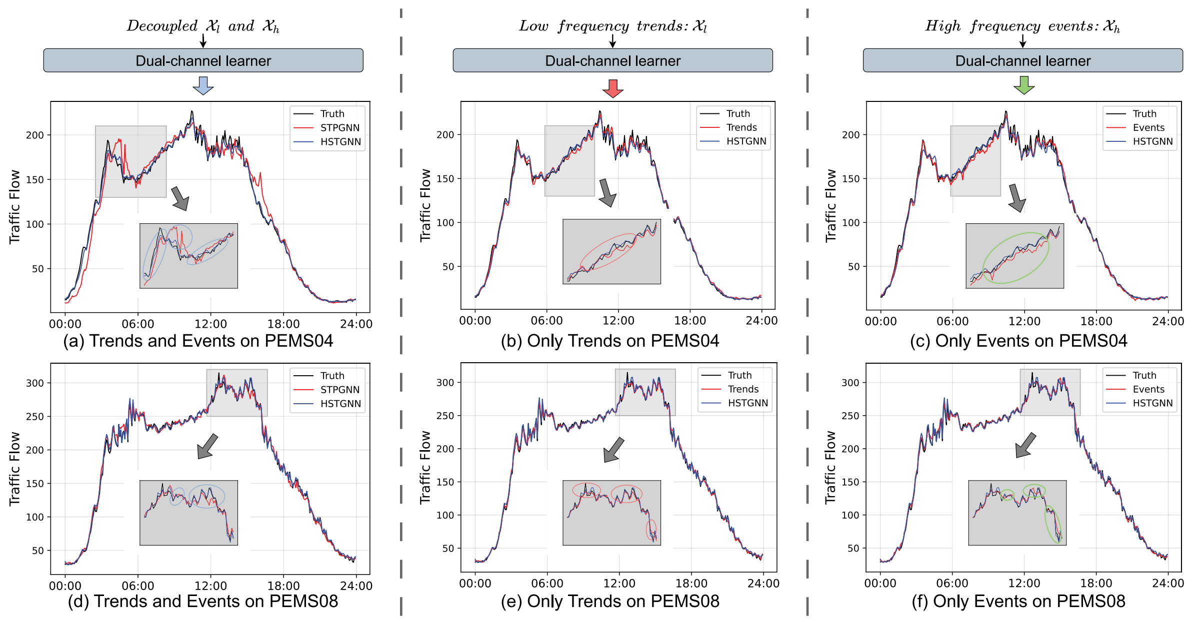

Fig. 8. Visual comparison of different input of HSTGNN's variants on PEMS04 and PEMS08 datasets.

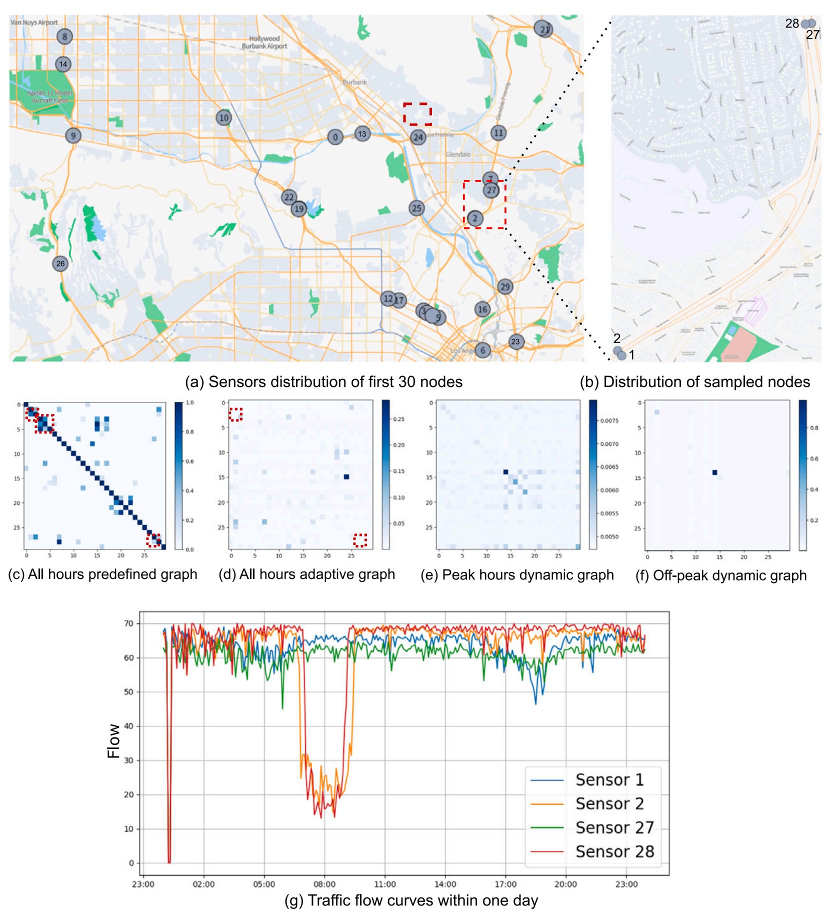

Fig. 9. (a) Sensors distribution of first 30 nodes in METR-LA dataset. (b) Distribution of sampled nodes. (c) All hours heatmap of the predefined graph. (d) All hours heatmap of the adaptive graph. (e) The heatmap of peak hours dynamic graph. (f) The heatmap of off-peak hours dynamic graph. (g) Traffic flow curves of sampled nodes.

the perspective of geographical relationships and traffic speed curves, the above cases demonstrate that the predefined adjacency matrix fails to reveal the true patterns of traffic information transmission. On the other hand, the adaptive graph learning algorithm provides a stable structure for the road network over the long term, while the dynamic graph aims to represent time-varying spatial correlations. Our hybrid graph learning module ensures that our model adequately captures the comprehensive spatial dependencies.

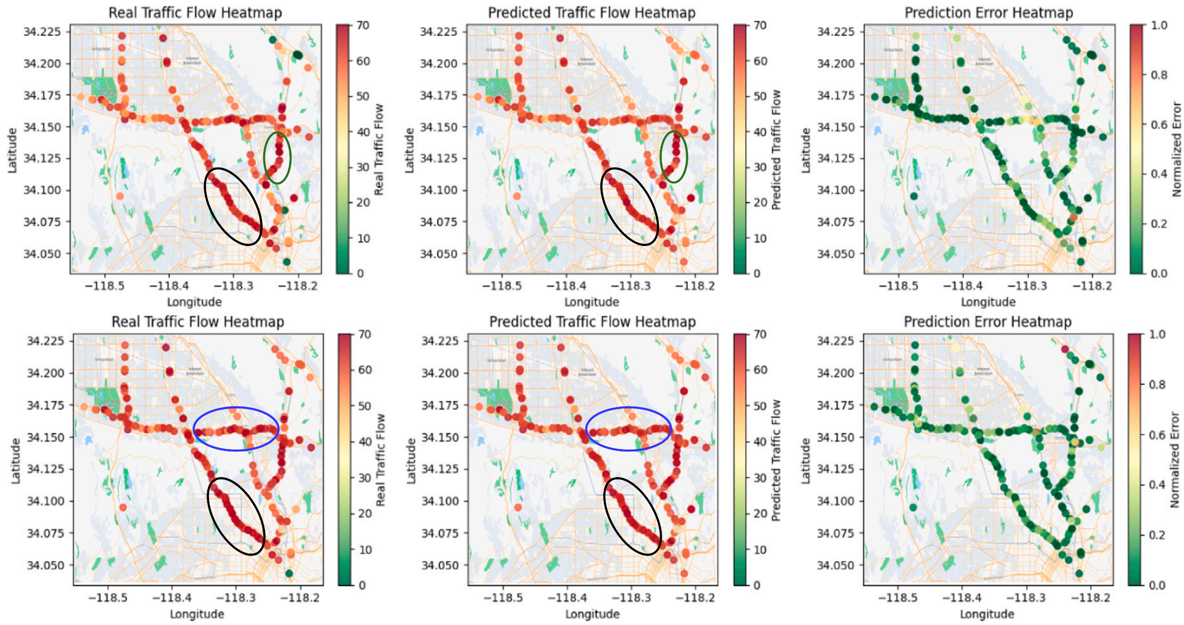

Fig. 10. Traffic flow forecasting visualization on METR-LA dataset. Black mark: Hollywood Freeway. Green mark: Glendale Freeway. Blue mark: Ventura Freeway. The prediction error is normalized by maximum value normalization.

5.10. *Case study (RQ7)*

To explore the performance of HSTGNN in real-world scenarios, we employ a heatmap visualization in Fig. 10. Specifically, we use three representative trunk streets, Hollywood Freeway, Glendale Freeway, and Ventura Freeway, in Los Angeles City. For this analysis, we randomly select two timestamps to demonstrate the model's performance. The results indicate that HSTGNN performs effectively in real-world scenarios, accurately capturing complex traffic patterns. By analyzing the heatmaps, we observe several notable traffic flow characteristics in the hotspot regions. For instance, on the Hollywood Freeway (marked in black), traffic flow is notably higher near major intersections, reflecting their importance as pivotal nodes in the road network. Meanwhile, the Ventura Freeway (marked in blue) exhibits varying traffic patterns across different segments, with some areas showing lower traffic flow, potentially indicating reduced congestion or increased use of alternative routes during certain periods.

Another interesting finding from the prediction error heatmap is that, HSTGNN can accurately predict traffic flow in the hotspot areas, highlighting its capability to model spatial–temporal variations. Such insights can further enhance the interpretability of the model and guide its application in optimizing traffic flow, mitigating congestion, and improving decision-making for intelligent transportation systems (ITS). By analyzing variations across specific segments and nodes, HSTGNN delivers valuable insights to urban planners and traffic managers, enabling them to tackle critical challenges in real-world traffic scenarios effectively.

6. Conclusion

In this paper, we introduce a novel deep learning model, HSTGNN, to address the traffic prediction problem. In the temporal dimension, we design a temporal decoupling layer to reveal the underlying and heterogeneous time patterns. Furthermore, a dual-channel learning module is designed to extract complementary information from different components. In the spatial dimension, we propose a dynamic spatial–temporal graph learning approach to capture time-varying spatial correlations. Moreover, considering the inherent properties of road networks, we further design a hybrid graph learning module to reflect compound spatial dependencies. Finally, extensive experiments on six real-world datasets are conducted to verify the effectiveness of HSTGNN. The experimental results show that our method outperforms existing state-of-the-art methods in terms of regular and long-term traffic forecasting tasks.

Future research will focus on three key directions: (1) investigating the intrinsic structure of traffic series to mitigate distributional biases and enhance prediction reliability; (2) developing advanced graph-based methodologies with pattern-aware diffusion mechanisms to optimize long-term dependency modeling and error resilience; (3) integrating contextual factors, including environmental and socioeconomic influences, to strengthen the model's adaptability and predictive performance.

CRediT authorship contribution statement

Peng Wang: Writing – review & editing, Validation, Funding acquisition, Conceptualization. **Longxi Feng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Yijie Zhu:** Writing – review & editing, Methodology. **Haopeng Wu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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